Printable Pattern

An Onshape custom feature that cuts hole patterns shaped to print without supports.

User guide · feature version of 2026-10-11
FeatureScript 3083 · units: mm

QUICK START

1. Start **Printable Pattern** and select one or more **faces or sketch regions**.
2. Choose the **pattern**, then **cell size**, **rib width** and **edge margin**.
3. Set the **print direction**: pick the plane or face that sits on the build plate.
4. Check **max overhang angle** and **max bridge width** against your printer.
5. Choose **Through all** or **Blind** and accept.
6. Read the blue message: it gives the cell count, the shape chosen and why. A yellow message means a forced style will not print cleanly.

> **Install:** paste **PatternMaker.fs** into a Feature Studio and commit. The feature then appears in the Part Studio toolbar under custom features.

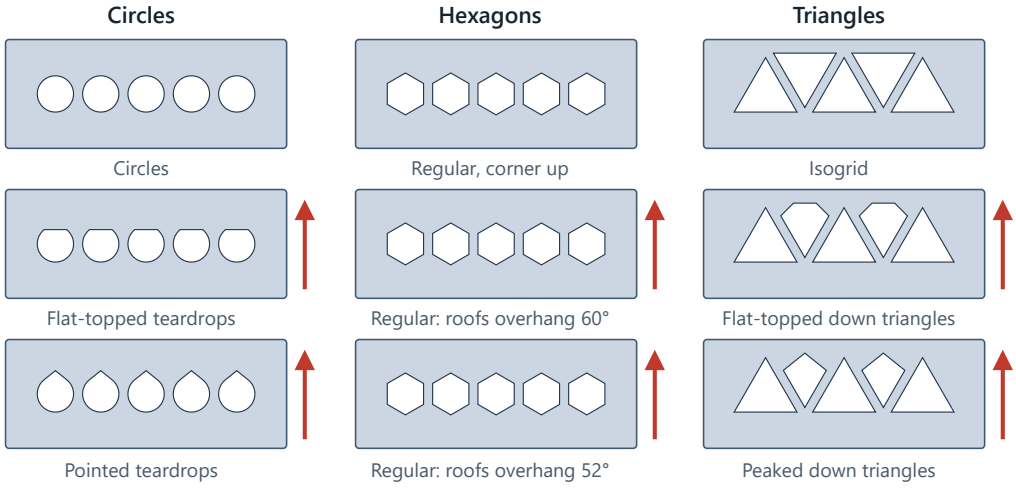
GOOD TO KNOW

- **Whole cells only.** A cell that would cross the region's edge is left out, so no hole is clipped.
- **Hexagons are regular, corner up.** On an upright wall their roofs overhang 60°; the message says so when that is over your limit. To stay within it, choose **Point up, stretched to print**, or **Flat top** to bridge.
- **No cells?** Reduce the cell size, rib width or edge margin.
- **Limit:** 4000 cells per region. Increase the cell size if you reach it.

SETTINGS

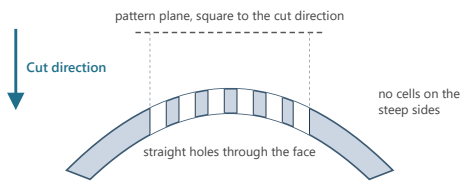
Faces or sketch regions	What to pattern; each selection is patterned on its own. Split a face first to pattern only part of it.
Pattern and style	Circles, hexagons or triangles. For circles and triangles, **Auto** picks a shape that prints. Hexagons are regular with a corner up unless you pick another style.
Tip angle	Hexagons only: turns the upper corner away from straight up, up to 30° either way.
Cell size	Diameter of the largest circle that fits in a hole (5 mm): a hexagon's width, a triangle's inscribed circle.
Rib width	Smallest wall left between holes (1.6 mm).
Edge margin	Smallest solid left between a hole and the region's edge (2 mm).
Print direction	Which way is up on the printer: a plane, planar face, straight edge or mate connector. Empty means +Z. The arrow flips it.
Max overhang angle	Measured from vertical, as slicers do (45°).
Max bridge width	Widest flat ceiling a hole may have (5 mm). 0 allows no bridges.
Pattern angle	Rotates the pattern. Used only when every orientation prints.
Cut direction	The hole axis. Empty means the face's own normal. Required for curved faces.
End type	**Through all**, or **Blind** with a depth. Cuts into the part; the arrow reverses it, **Both sides** cuts both ways.
Parts to cut	Empty means the face's own part (every part, for sketch regions).
Keep pattern sketch	Leaves the cell outlines behind as a sketch.

HOW THE HOLES ADAPT TO THE PRINT DIRECTION



Red arrow = print up. Pictures are drawn by the feature's own math: cell 8 mm, rib 2 mm, edge margin 2 mm, max overhang 45°, max bridge 5 mm.

CURVED FACES



Select the curved face, then set **Cut direction** to a plane, planar face or straight edge. The holes go straight through along that direction.

Which way the direction points does not matter: the cut goes into the part.

- **Through all** cuts everything behind the face. On a tube, that is both walls.
- **Blind** makes pockets whose floor follows the face: the depth is measured square to the surface.
- Cells keep the edge margin from the face's edges and from where the face curves out of view.